

Q-1: When is It Warmer?

Given an array of integers `temperatures` representing the daily temperatures, return an array `answer` such that `answer[i]` is the number of days you have to wait after the i th day to get a warmer temperature. If there is no future day for which this is possible, set `answer[i] = 0`.

Example Test Cases:

TEST CASE 1Input: `temperatures = [73,74,75,71,69,72,76,73]`Output: `[1,1,4,2,1,1,0,0]`**TEST CASE 2**Input: `temperatures = [30,40,50,60]`Output: `[1,1,1,0]`**TEST CASE 3**Input: `temperatures = [30,60,90]`Output: `[1,1,0]`

Constraints:

- $1 \leq \text{temperatures.length} \leq 10^5$
- $1 \leq \text{temperatures}[i] \leq 10^9$

Expected Time Complexity: $\Theta(N)$

Input Format:

1. An array of integers `temperatures`, where each element `temperatures[i]` represents the temperature on the i -th day.

Output Format:

1. Return an array `answer` such that `answer[i]` is the number of days you have to wait after the i -th day to get a warmer temperature. If there is no future day for which this is possible, keep `answer[i] = 0` instead.

Q-2:K-Node Flip

Given the head of a linked list, reverse the nodes of the list k at a time, and return the modified list.

k is a positive integer and is less than or equal to the length of the linked list. **If the number of nodes is not a multiple of k , then the remaining nodes should remain as they are.**

Note: You may not alter the values in the list's nodes, only the nodes themselves may be changed. You must solve the problem in $O(1)$ **extra memory space**.

Example Test Cases:

TEST CASE 1

Input: head = [1,2,3,4,5], $k = 2$

Output: [2,1,4,3,5]

TEST CASE 2

Input: head = [1,2,3,4,5], $k = 3$

Output: [3,2,1,4,5]

Constraints:

- The number of nodes in the list is n .
- $1 \leq k \leq n \leq 10^9$.
- $0 \leq \text{Node.val} \leq 10^9$.

Expected Time Complexity: $\Theta(N)$

Input Format:

1. The input consists of two elements:
 - (a) A singly linked list where each node contains an integer value.
 - (b) An integer k representing the group size for reversal.

Output Format:

1. Return the head of the modified linked list after reversing every k nodes.

Instructions: You have been provided with a template file containing a `main` function that takes the input and calls the `solve` function for both questions. The first line of input for each question contains `T` –the number of test cases. The `solve` function is called separately for each test case. Your task is to implement the `solve` function for each question.

The function must:

- For (When is It Warmer?): Return a vector of size `n`, containing the number of days you have to wait after the `-th` day to get a warmer temperature. If no such future day exists, return `.`
- For (K-Node Flip): Return a linked list where the nodes are reversed in groups of `k`.